Exam 1

Vectors

1. What is a vector? What are its components, its length (or norm), and its direction?

2. Add, subtract, and scale vectors algebraically. Describe and carry out these operations geometrically.

3. Find a vector with a specified direction and length (including lying along a line).

4. Compute the dot product, the cross product, and the scalar triple product. What are their geometric interpretations? What happens if you change the order of the inputs?

5. Compute the angle between two nonzero vectors. Is it acute or obtuse, or are the vectors perpendicular?

6. Find a unit vector perpendicular to two given vectors, including the case when the given vectors are parallel.

7. Find the area of a triangle or parallelogram, or the volume of a parallelepiped, using vectors.

8. Determine whether three vectors are coplanar.

9. Use vector identities to simplify expressions without computing components. For example, express $\mathbf{u} \cdot \Big(\mathbf{v} \times (\mathbf{u} + \mathbf{v} + \mathbf{w})\Big)$ in terms of $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$, and explain which properties you used.

10. Compute and sketch scalar and vector projections; decompose a vector into parallel and perpendicular components.

Lines and planes

11. Write a parameterization of a line from points or geometric conditions.

12. Write a plane's equation from points, normal vectors, or contained lines.

13. Determine whether two lines coincide, are parallel, intersect, or are skew; find an intersection when one exists.

14. Find a line's intersection with a plane, including recognizing parallel or contained cases.

15. Parameterize the intersection of two planes.

16. Find angles between lines, between planes, and between a line and a plane.

17. Find the distance from a point to a line and identify the closest point.

18. Find the distance from a point to a plane and identify its orthogonal projection.

Parametric curves and motion

19. Sketch or identify the curve traced by a vector-valued function, including its orientation.

20. Construct a curve lying on a given surface and joining specified endpoints. Verify that a given curve lies in a specified plane or on another specified surface.

21. Parameterize a circle, ellipse, or curve formed by intersecting surfaces. Choose a parameter interval that traces the desired curve exactly once.

22. Differentiate and integrate vector-valued functions component by component.

23. State and apply the differentiation rules for dot products, cross products, and products of scalar-valued and vector-valued functions.

24. Find and sketch a tangent vector, unit tangent vector, and tangent line at a specified point.

25. Find where curves intersect and the angle between their tangent lines.

26. Find where a curve's tangent satisfies a geometric condition, such as being parallel to a plane.

27. Compare two parameterizations of the same curve, including their directions and speeds. Explain why a regular one-to-one reparameterization leaves arc length unchanged.

28. Find velocity, speed, and acceleration from position. Recover velocity and position from acceleration or velocity and initial conditions.

29. Calculate displacement, straight-line distance between endpoints, and total distance traveled (arc length), setting up and evaluating the arc-length integral when possible. What is the difference between distance and displacement?

Functions of several variables and partial derivatives

30. Find and sketch a function's domain, including its boundary; find its range and justify that all claimed values occur. Find and sketch traces, level curves, or level surfaces.

31. Match a formula to a surface or contour plot and explain the match.

32. Compute first and higher-order partial derivatives.

33. Interpret a partial derivative's sign and units in context.

34. Estimate a partial derivative from a table of values.

35. Determine signs of first and second partial derivatives from a contour plot. Locate a point satisfying specified derivative conditions.

36. Explain when mixed partial derivatives are equal and verify their equality for a given function.

Chain rule and implicit differentiation

37. Write down the multivariate chain rule for $f(x, y)$ where $x = x(u, v)$ and $y = y(u, v)$, giving the derivatives with respect to both u and v. Explain intuitively why the rule is correct.

38. Find partial derivatives of a composition involving multiple intermediate variables. Apply product and quotient rules as needed, including for functions defined using dot products, cross products, and products of scalar-valued and vector-valued functions.

39. Use given partial derivatives and input rates to find an output's rate of change. Find an input rate needed to keep an output constant.

40. Find partial derivatives of a variable defined implicitly by an equation.

Bonus Challenge Problems

41. Find the distance between parallel planes.

42. Project a line onto a plane.

43. Explain the relationship between constant speed and perpendicular velocity and acceleration.